

Angles Between Vectors

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Definition. Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^m$ be non-zero vectors. The angle between them is the unique α between 0 and π such that

$$\cos \alpha = \frac{\mathbf{v}}{\|\mathbf{v}\|} \cdot \frac{\mathbf{w}}{\|\mathbf{w}\|}.$$

Proof (Geometric Agreement). We want to show now that this definition agrees with our intuition for two dimensions. Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^m$ be non-zero vectors. We observe that the angle between them is independent of their lengths (as long as it is non-zero). Thus, we consider the unit vectors associated with $\mathbf{v}$ and $\mathbf{w}$,

$$\hat{\mathbf{v}} := \frac{\mathbf{v}}{\|\mathbf{v}\|} \quad \text{and} \quad \hat{\mathbf{w}} := \frac{\mathbf{w}}{\|\mathbf{w}\|}.$$

Since the angle is also independent of the rotation of the two vectors, we can align $\hat{\mathbf{v}}$ with the vector $\mathbf{e}_1$. We can assume w.l.o.g. that $\hat{\mathbf{w}}$ is to the left of $\hat{\mathbf{v}}$. We assume for now that α is acute. The case where α is obtuse is analogous.

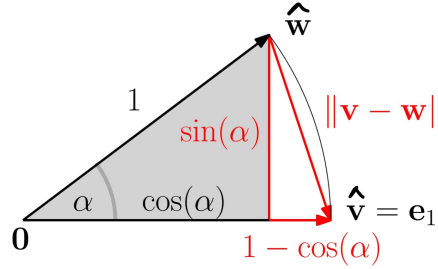


Figure 1: Illustration for acute α

We use the definitions of sin and cos in right triangles to find the side lengths of the legs of the red triangle and the gray triangle. Using the Pythagorean theorem on the red triangle, we get

$$\|\hat{\mathbf{v}} - \hat{\mathbf{w}}\|^2 = \sin^2 \alpha + (1 - \cos \alpha)^2 = 2 - 2 \cos \alpha.$$

We also know that

$$\begin{aligned} \|\hat{\mathbf{v}} - \hat{\mathbf{w}}\|^2 &= (\hat{\mathbf{v}} - \hat{\mathbf{w}}) \cdot (\hat{\mathbf{v}} - \hat{\mathbf{w}}) && \text{(Definition of norm)} \\ &= \hat{\mathbf{v}} \cdot \hat{\mathbf{v}} - 2\hat{\mathbf{v}} \cdot \hat{\mathbf{w}} + \hat{\mathbf{w}} \cdot \hat{\mathbf{w}} && \text{(Distributivity)} \\ &= \|\hat{\mathbf{v}}\|^2 + \|\hat{\mathbf{w}}\|^2 - 2\hat{\mathbf{v}} \cdot \hat{\mathbf{w}} \\ &= 2 - 2\hat{\mathbf{v}} \cdot \hat{\mathbf{w}} && \text{(Unit vectors)} \end{aligned}$$

Thus, we get

$$2 - 2 \cos \alpha = 2 - 2\hat{\mathbf{v}} \cdot \hat{\mathbf{w}},$$

which simplifies to

$$\cos \alpha = \hat{\mathbf{v}} \cdot \hat{\mathbf{w}} = \frac{\mathbf{v}}{\|\mathbf{v}\|} \cdot \frac{\mathbf{w}}{\|\mathbf{w}\|},$$

which agrees exactly with the above definition. $\square$

Proof (Well-Definedness). Since we want to use the above definition to *define* the angle between two vectors in higher dimensions, we must show that our definition is well-defined, i.e. that each pair of non-zero vectors $\mathbf{v}$, $\mathbf{w}$ is assigned a unique angle in $[0, \pi]$. By Cauchy-Schwarz, we know that

$$|\mathbf{v} \cdot \mathbf{w}| \leq \|\mathbf{v}\| \|\mathbf{w}\| \implies \left| \frac{\mathbf{v}}{\|\mathbf{v}\|} \cdot \frac{\mathbf{w}}{\|\mathbf{w}\|} \right| \implies -1 \leq \frac{\mathbf{v}}{\|\mathbf{v}\|} \cdot \frac{\mathbf{w}}{\|\mathbf{w}\|} \leq 1.$$

Since $\arccos$ is bijective on $[-1, 1]$, the angle is well-defined. □

Projective Property

By rearranging the above definition, we obtain a fundamental relationship between the scalar product and the angle between the involved vectors, namely

$$\boxed{\mathbf{v} \cdot \mathbf{w} = \cos \alpha \cdot \|\mathbf{v}\| \cdot \|\mathbf{w}\|}.$$

This is a remarkable property of the scalar product, called its *projective property*. This name is justified since for unit vectors and an acute angle α , this implies

$$\hat{\mathbf{v}} \cdot \hat{\mathbf{w}} = \cos \alpha,$$

meaning that the scalar product gives the length of the projection of $\hat{\mathbf{w}}$ to $\hat{\mathbf{v}}$ (cf. Figure 1). We will explore this property a bit more in the second part of the course, when talking about projections of vectors to subspaces of a general $\mathbb{R}^m$.